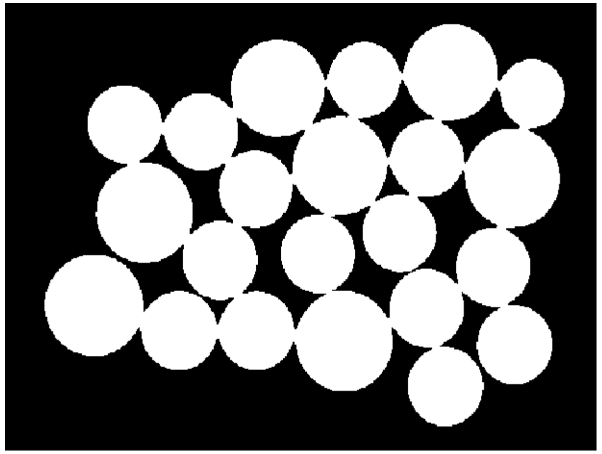
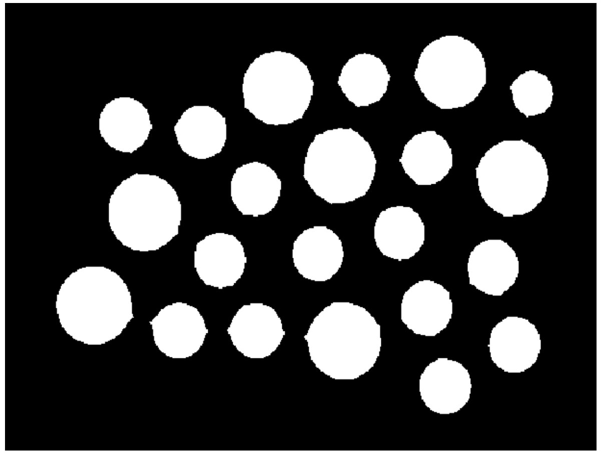
Morphological Image Processing

Application

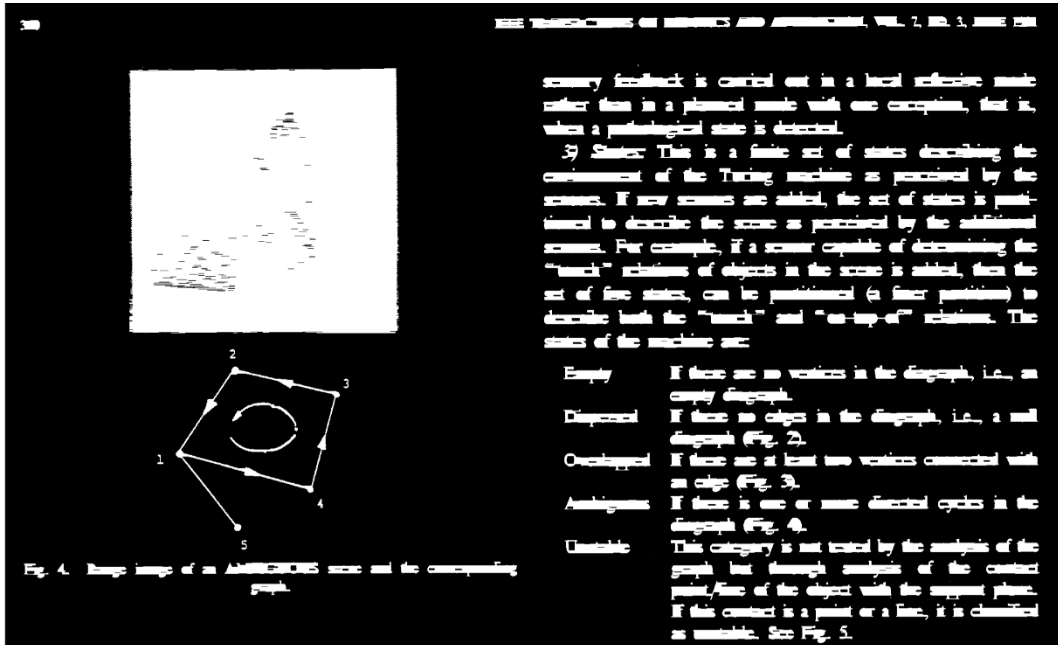
- Erosion

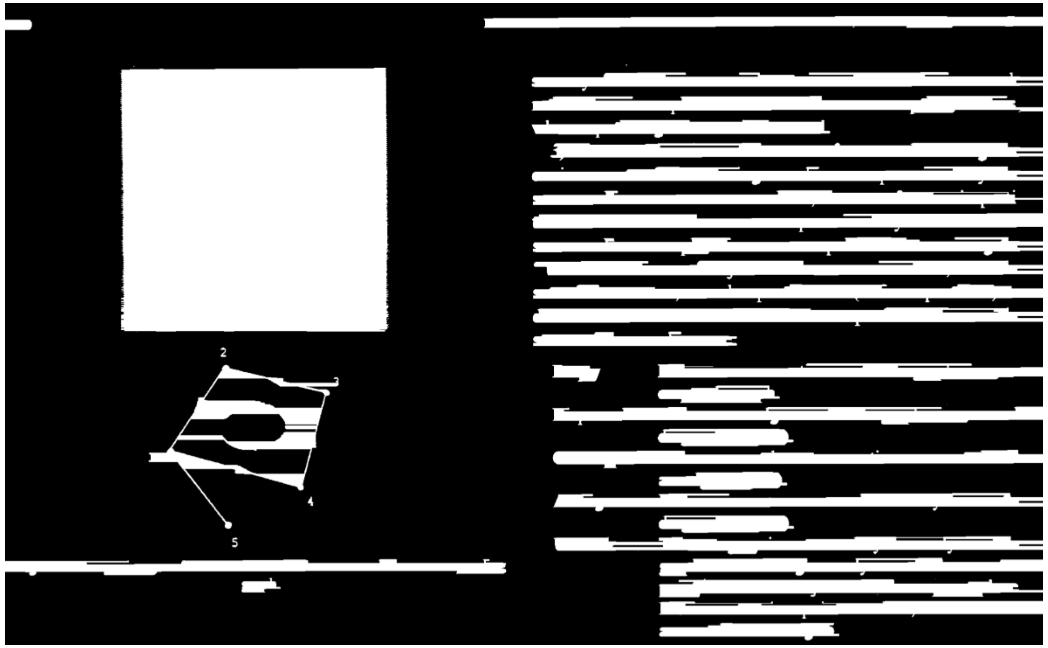




Application

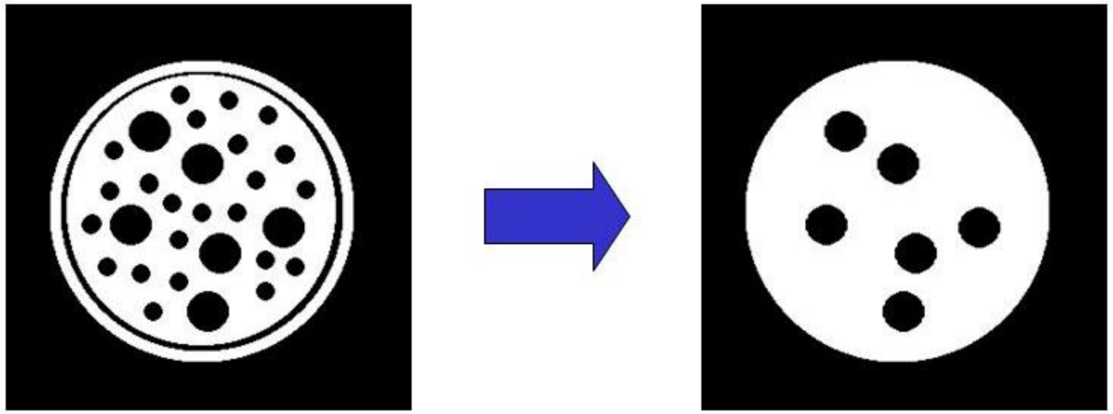
- Dilation





Morphological Operation

- Non-linear operations related to the shape or morphology of features in an image
- On Binary Images

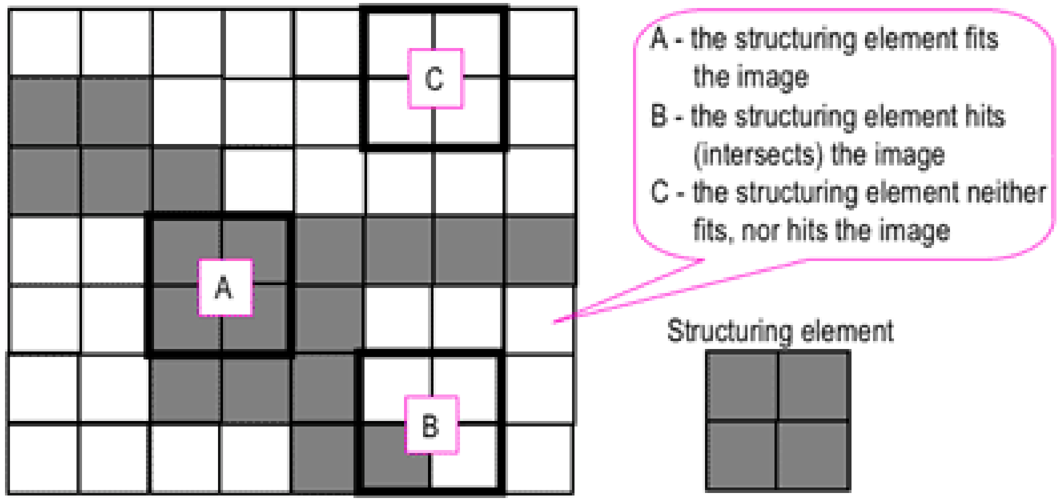


Structuring Element

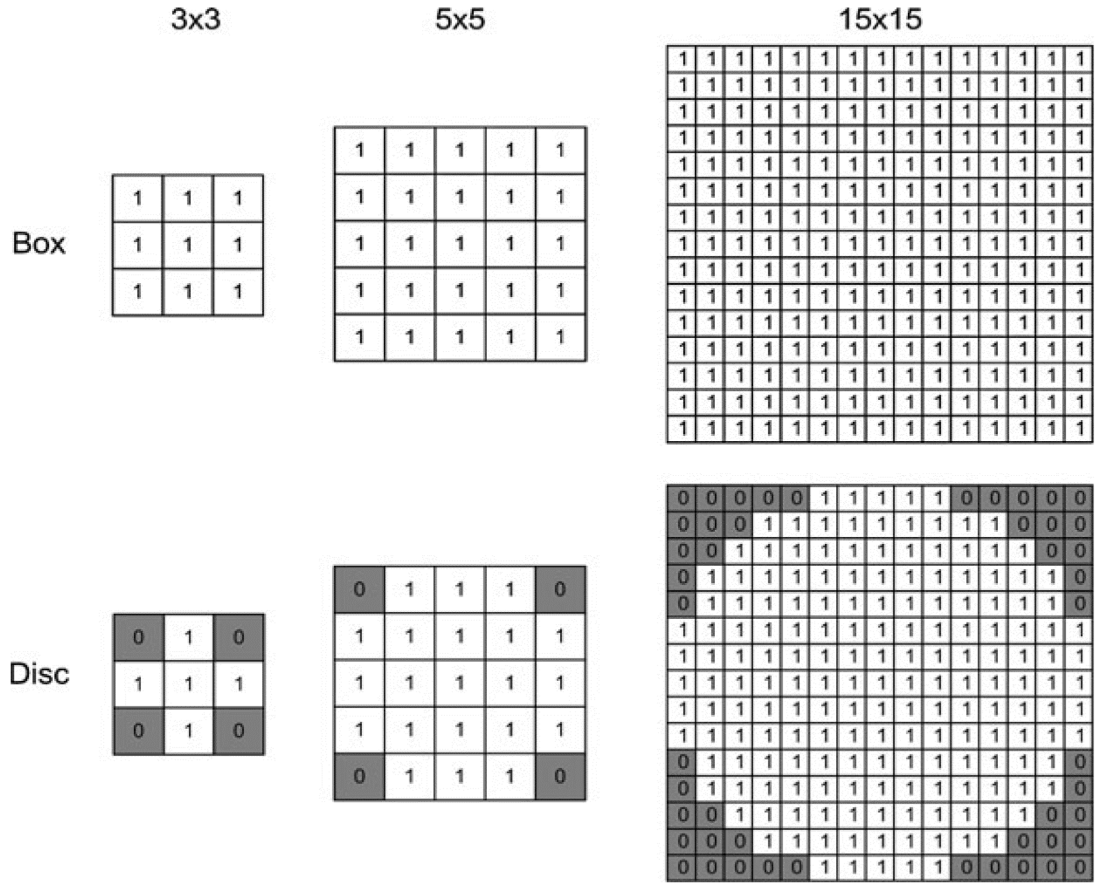
- A small shape or template called a **structuring element**, a matrix that identifies the pixel in the image being processed and defines the neighborhood used in the processing.

Structuring Element

- Hit, Fit or Miss



Different Types of Structuring Element



Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1

Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1 0

Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1 0 1

Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1 0 1 1

Example for Dilation

Input image   1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image   1 0 1 1 1

Example for Dilation

Input image   1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image   1 0 1 1 1 1

Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1 0 1 1 1 1 1

Example for Dilation

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 1 0 1 1 1 1 1 1

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0 0 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0 0 0 1

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0 0 0 1 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

1	1	1

Output Image 0 0 0 0 1 0 0

Example for 1D Erosion

Input image 1 0 0 0 1 1 1 0 1 1

Structuring Element

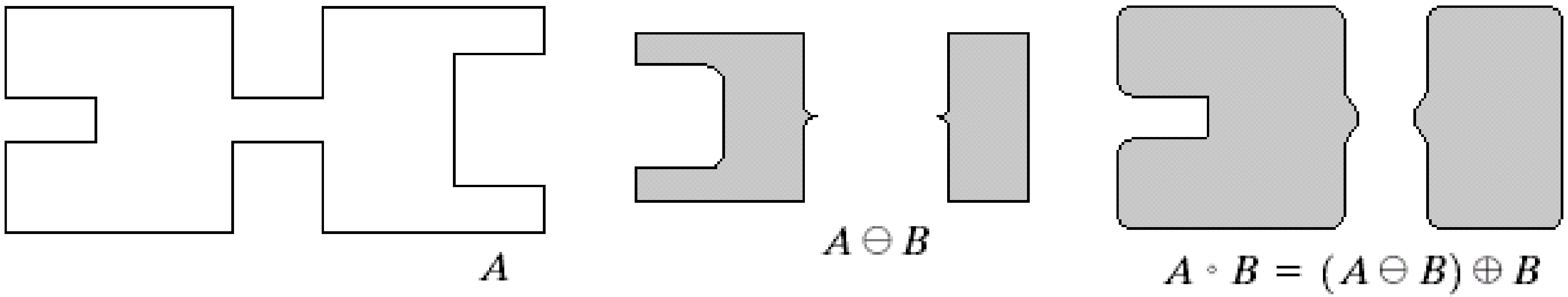
1	1	1

Output Image 0 0 0 0 1 0 0 0

Opening

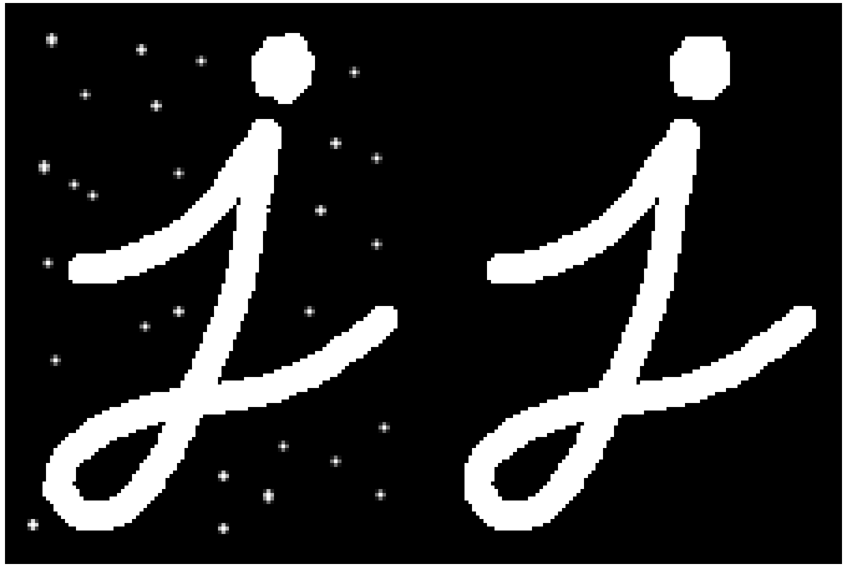
The opening of image *A* by structuring element *B*, denoted by *A*○B is simply an erosion followed by a dilation

$$A \circ B = (A \ominus B) \oplus B$$



Original shape After erosion After dilation (opening)

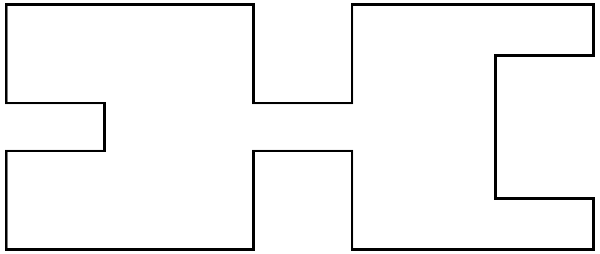
Opening



Closing

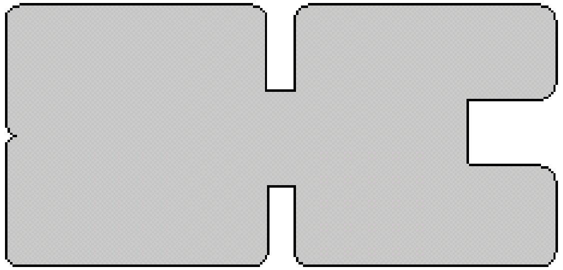
The closing of image *f* by structuring element *s*, denoted by *f* • *s* is simply a dilation followed by an erosion

$$A \cdot B = (A \oplus B) \ominus B$$



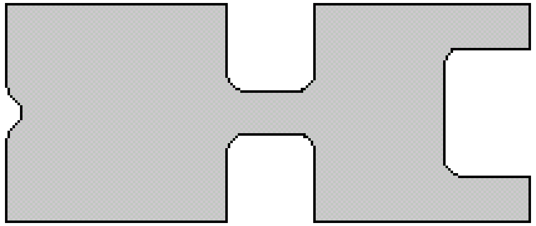
A

Original shape



$A \oplus B$

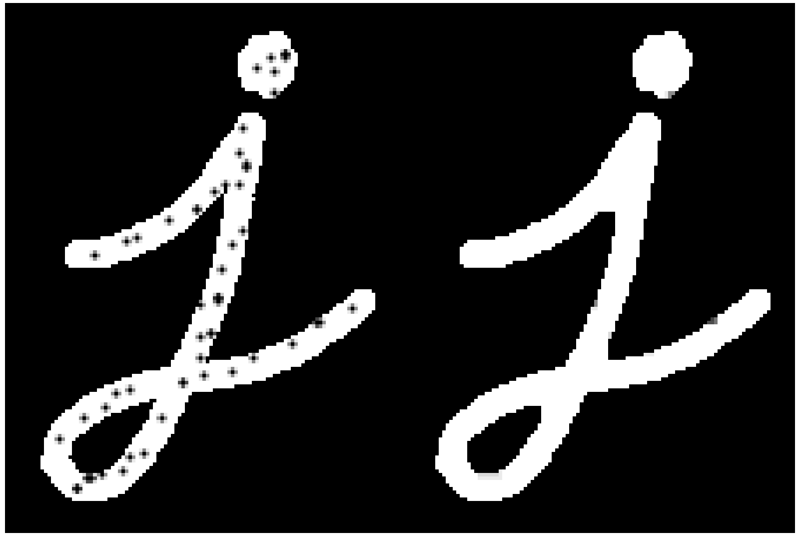
After dilation



$A \cdot B = (A \oplus B) \ominus B$

After erosion (closing)

Closing



Exercise: Which one should we apply here?